

MASHNUNUL HUQ

Address: Ben-Gurion-Ring 50, 60437, Frankfurt am Main | LinkedIn: <https://www.linkedin.com/in/mashnunul-huq-javed-53715a110/>
Tel.: +49-17634646865 | E-Mail: mashnunul.huq@stud.fra-uas.de | Portfolio Website: <https://conundrumsolutions.pythonanywhere.com/>
GitHub Repo: <https://github.com/javedcoding>

EDUCATION

Information Technology (M.Eng.) Frankfurt University of Applied Sciences Deutschland	10/2021 – 12/2025 (Tentative) (Current CGPA: 2.1)
Electrical and Electronic Engineering (B.Sc.) Ahsanullah University of Science and Technology Bangladesh	10/2013 – 12/2017 (Final Grade: 1.9)

WORK EXPERIENCE

Student Assistant in Computational Intelligence Lab (Prof. Dr. Andreas Pech) Frankfurt University of Applied Sciences Deutschland	Since 06/2023
- Design of data pipelines for processing and storing large spectrogram data using Azure CosmosDB and DataLakeHouse with ETL pipelines for training LLM (NLP) with 98% cosine similarity. - Implementation of TCP/ZeroMQ connection between I2C sensor coding in C and client ultrasonic data acquisition in Python PyQt6 of Redpitaya Embedded System for autonomous driving escalating memory for 75k data points. [Link]	

UX-Specialist and Data Analyst Adiva Graphics Bangladesh	06/2019 – 06/2020
- Automated creation of meaningful charts of clinical trial data for medication adherence using Tableau, Python Matplotlib and Django, reducing reporting time by 40%. - Automated data collection pipeline with Python programming, MySQL and PostgreSQL from the data pool, reducing data collection time by 10%. Use of KNN, regression and PCA for ML training.	

DevOPS and Data System Engineer Dulal Brothers Limited Kashimpur, Bangladesh	07/2018 – 05/2019
- Automation of a dye mixing system with C++ and MySQL to reduce production time by 30%. - Created software for automatic color weighting Program for automatic color dispensing with Python Programming language for sensor fusion.	

Software Internship CodeSmith Tech Ltd. Dhaka, Bangladesh	12/2017 – 05/2018
Designing Android APP with Flutter.io and website and ERP-System building with Django.	

IT - SKILLS

Programming Languages: Python, C++
Framework: TensorFlow, PyTorch, ScikitLearn, Flake8
Database Systems: MySQL, PostgreSQL, MongoDB
Technology: Deep Learning, Microsoft Azure Cloud, Generative KI, Git-Hub, Tableau
OS-System: Windows, Linux
Libraries: Numpy, Pandas, Keras, Fairseq, Langchain, scipy

CERTIFICATION

Microsoft Certified: Azure Data Scientist Associate (04/2024)
Microsoft Certified: Azure AI Fundamentals (01/2024)
HashiCorp Certified: Terraform Associate - Hands-On Labs (07/2024),
Udemy
SQL-MySQL for Data Analytics and Business Intelligence (12/2023),
Udemy
Deep Learning Specialization (08/2023), Deeplearning.ai
Generative AI with Large Language Models (06/2024), Coursera

LANGUAGES

Deutsch - B2
English - C1
Bengalisch – Mother Tongue

PROJECTS

Person Detection with SONAR CNN und MobileNetV2

Frankfurt University of Applied Sciences

11/2021 – 12/2022

Supervision: Prof. Dr. Andreas Pech

Technologies: Redpitaya Embedded System, Ultrasonic Sensor, PyQt6, I2C Protocol, Transfer Learning, MobileNetv2

- 96% accuracy in human presence detection using a low-cost SONAR sensor and RedPitaya Embedded system with arm processor, comparing customized CNN and MobileNetv2 models with t-SNE's Neighbourhood similarity graph.
- Design and implementation of data pipelines to process and store large spectrogram data using Azure CosmosDB and Data Lake House with ETL pipelines for training CNN.

Video Learning with NeocortexAPI and FFmpeg using Microsoft Azure

Frankfurt University of Applied Sciences

11/2021 – 12/2022

Supervision: Dr. Damir Dobric and Prof. Dr. Andreas Pech

Technologies: Machine Intelligence, HTM (Hierarchical Temporal Memory), Deep learning(NN), C#, DotNet, Python, Microsoft Azure.

- Implementation of FFMPEG and EMGUCV for video slicing and image framing, training in Azure Cloud using blob storage, C# .Net 5 with 95% accuracy in frame prediction.
- Development of an ETL pipeline with Azure Blob Storage for uploading video files and generating frames with GitHub CI/CD pipeline.

Team Leader in the development of an Authentication Server

Frankfurt University of Applied Sciences

10/2023 – 01/2024

Supervision: Dr. Patrick Wacht

Technologies: Python, Django, Agile Development, Kanban Board

- By leading a team building an authentication server on Django, 80% more success was achieved than other teams, reducing the project duration by 2 weeks.
- Implemented two authentication servers with JWT and OAuth2 using agile development principles.

“Kidnapped Robot”– Problem Solving with ROS, C and Python

Frankfurt University of Applied Sciences

11/2022 – 02/2023

Supervision: Prof. Dr. Peter Nauth

Technologies: Machine Intelligence, ROS (Robot Operating System), C++, RPLIDAR, Intel Real Sense 3D Camera

- Solving the “Kidnaped robot problem” in ROS using computer vision (YOLOv7) with the help of RPLIDAR and depth camera with Python and C, reducing the area detection delay by 25%.
- Creating a Gazibo simulation to test the robot position detection in different scenarios with later implementation in the real turtle bot and achieving 10% faster detection than the previously installed system.

Prof. Dr. Andreas Pech
Frankfurt University of Applied Sciences
Fachbereich 2: Informatik und Ingenieurwissenschaften
60318 Frankfurt am Main

Frankfurt University of Applied Sciences (FRA-UAS)
Nibelungenplatz 1 • 60318 Frankfurt am Main • Germany

Empfehlungsschreiben

Herr BSc. (Electrical and Electronic Engineering) Mashnunul Huq, geb. am 29. Juli 1995 in Dhaka (Bangladesh), ist mir durch seine Teilnahme an meinen Lehrveranstaltungen „Software Engineering“, „Cloud Computing“, „Stochastic Signals and Systems“, „Computational Intelligence“ und „Machine Learning“ als Student des Master-Studiengangs Information Technology bekannt.

Herr Huq ist im Labor für Computational Intelligence seit mehreren Semestern als Tutor tätig. Im Einzelnen:

Zeitraum	Umfang
01.06.2023 - 31.07.2023	40 Stunden pro Monat
Seit 01.10.2023	80 Stunden pro Monat

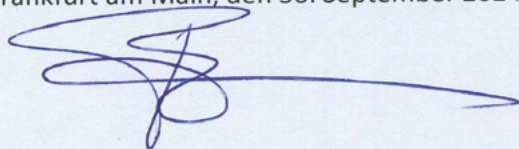
Seine Tätigkeit besteht in der Unterstützung der Studierenden in meinen Lehrveranstaltungen „Machine Learning“ und „Stochastic Signals and Systems“, sowohl bei der Vor- und Nachbereitung von Übungen als auch im Peer-to-Peer-Teaching. Für unsere Messapparaturen ist Software sowohl für die Datenakquisition, die Datenübertragung und das Trainieren von KI-Algorithmen zu erstellen. Wir verwenden ARM-Prozessoren in einer LINUX-Umgebung, die mit Windows-Rechnern kommunizieren, welche für Machine Learning eingesetzt werden. Herr Huq hat die heterogene Umgebung betreut und Programme in Python, C und Matlab erstellt.

Er hat seine Aufgaben stets sehr gründlich, gewissenhaft, sorgfältig, sowie zur vollsten Zufriedenheit erfüllt.

Herr Huq ist ein überdurchschnittlicher Tutor, der aufgrund seines großen Engagements bei den Studierenden sehr beliebt ist. Sein Verhalten gegenüber Laboringenieuren und Vorgesetzten war stets einwandfrei.

Dieses Empfehlungsschreiben wurde zur Vorlage bei Bewerbungen erstellt. Ich wünsche Herrn Huq bei seiner Bewerbung viel Glück.

Frankfurt am Main, den 30. September 2024



Prof. Dr. Andreas Pech
Leiter des Labors für Computational Intelligence

Frankfurt University of Applied Sciences
Fb 2: Informatik und Ingenieurwissenschaften
Computer Science and Engineering

Nibelungenplatz 1 - Telefon: 069 1533-0
60318 Frankfurt am Main